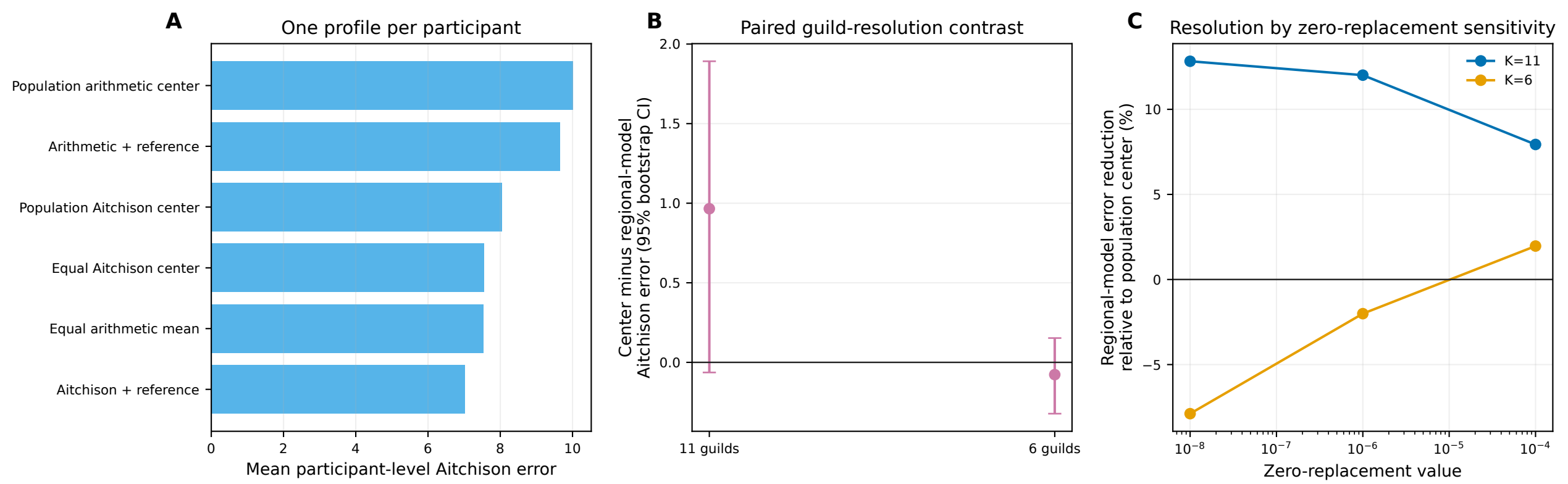
A
One profile per participant
Population arithmetic center
Arithmetic + reference
Population Aitchison center
Equal Aitchison center
Equal arithmetic mean
Aitchison + reference
0
2
4
6
8
10
Mean participant-level Aitchison error
B
Paired guild-resolution contrast
Center minus regional-model
Aitchison error (95% bootstrap CI)
2.0
1.5
1.0
0.5
0.0
11 guilds
6 guilds
C
Resolution by zero-replacement sensitivity
K=11
K=6
Regional-model error reduction
relative to population center (%)
10
5
0
−5
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}
Zero-replacement value